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* Available on NDL's web site: http://www.ars.usda.gov/nutrientdata.

Notes on Foods

Introduction

The information contained in ”Notes on Foods” was initially published in printed copies of Agriculture Handbook No. 8 (AH-8), which were presented as individual sections by food groups. In addition to a description of the tables and how nutrient values were determined, each food group section included a portion, called “Notes on Foods” with information specific to each food group. The information on the database, nutrient values and formats has been published separately as the documentation accompanying each release of the USDA National Nutrient Database for Standard Reference (SR; NDL, 2013). At this time, “Notes on Foods” are included in this document for only some of the sections previously available in the printed copies. It is anticipated that this document will expand, as information for the remaining food groups is added.

Data are obtained from a variety of sources (Figure 2). These include the scientific literature, data provided by food companies and trade associations, other government agencies and USDA-sponsored contracts. In a number of cases, various trade associations have worked with the Nutrient Data Laboratory (NDL) to design analytical studies to obtain new data on various food items. These studies are described in greater detail in their respective sections below. Since 1997, USDA-sponsored contracts have been conducted under the aegis of the National Food and Nutrient Analysis Program (NFNAP), which is described below.

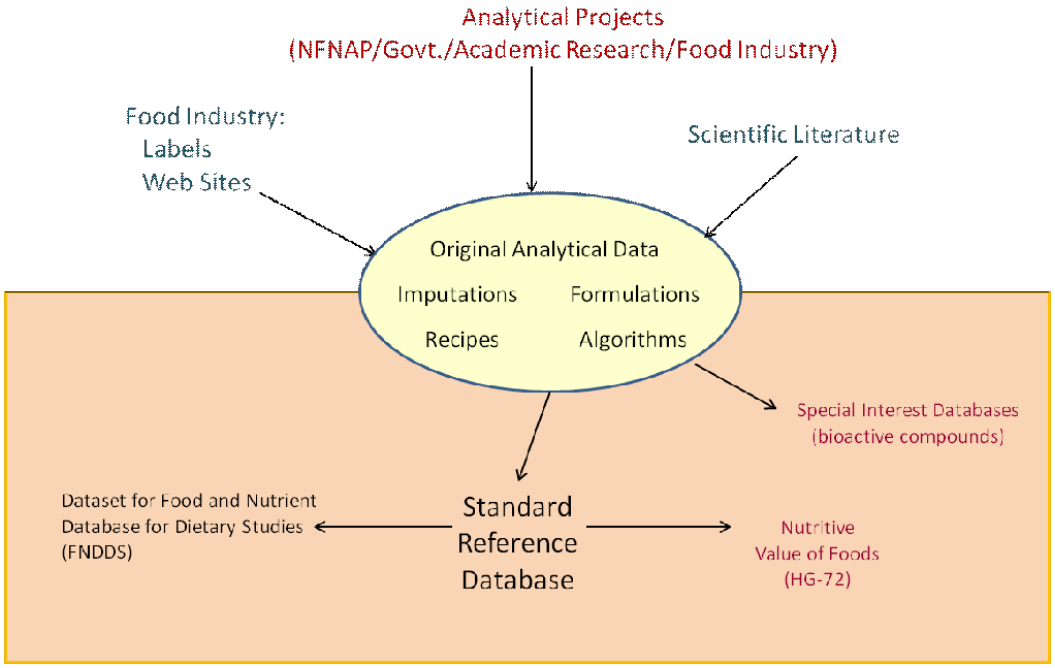


Figure 2. Sources of data and NDL products

National Food and Nutrient Analysis Program

In 1997, the NDL, in cooperation with the National Heart Lung and Blood Institute and other Institutes and Offices of the National Institutes of Health (NIH), instituted the National Food and Nutrient Analysis Program. In 2005, The National Cancer Institute took over the lead role in coordinating the program at NIH. The goals of NFNAP are to improve the quantity and quality of data in the USDA National Nutrient Databank which has resulted in annual updates of the USDA National Nutrient Database for Standard Reference (USDA, 2014) and a number of Special Interest Databases: isoflavones (USDA, 2008), choline (USDA, 2004a), proanthocyanidins (USDA, 2004b), fluoride (USDA, 2005), and flavonoids (USDA, 2013). To achieve these goals, five principle aims were established:

1. Identify and rank foods and nutrients for analysis;
2. Evaluate existing data for foods and nutrients;
3. Develop strategies for sampling;
4. Process and analyze foods; and
5. Review and disseminate results.

Since its inception in 1997, over 1,900 unique food items have been sampled and analyzed under NFNAP—some of these have been sampled and analyzed more than once as products and formulations have changed. To date, values for over 1,600 of these food items have been incorporated into SR. The process of acquiring, evaluating, and disseminating food composition data is continuous. At any time, new samples are being collected, prepared and analyzed and data for samples already analyzed are being revised and processed through NDL's Nutrient Data Bank System (NDBS). Details of these studies are described in specific chapters on each food group, which follow this section. A number of trade associations in the food industry, such as the National Cattleman's Beef Association, the National Pork Board, the Produce for Better Health Foundation, the Mushroom Council, the American Egg Board and others have worked with NDL to analyze food items in their product category sectors, using protocols adapted as part of NFNAP. Details of each of these studies are described in the specific chapter for each food group.

Identify Key Foods and critical nutrients for sampling and analysis

To identify and rank foods and nutrients for analysis, the Key Foods approach (Haytowitz *et al*., 2000; Haytowitz *et al*., 2002) was used. Key Foods are those foods which in aggregate contribute 75% of the nutrient intake for selected nutrients of public health importance from the diet. The most current Key Foods list was generated using weighted two-day food consumption data from the National Health and Nutrition Examination Survey (NHANES) 2007-08 Data Files (NCHS, 2010) and food composition data from SR22 (issued in 2009). For the current Key Foods list, targeted nutrients (total fat, food energy, total sugar, total dietary fiber, calcium, iron, potassium, sodium, β-carotene, α-tocopherol, vitamin C, vitamin B_{12}, choline, cholesterol and saturated fatty acids) were those identified in the Dietary Guidelines Advisory

Committee Report on the *Dietary Guidelines for Americans*, 2010 (DGAC, 2010) and the *Dietary Guidelines for Americans*, 2010 (USDA & USDHHS, 2010) as "shortfall" nutrients (limited in the diet) or nutrients of excess consumption, in particular those associated with poor health outcomes. Other nutrients of concern such as *trans* fatty acids were considered but not included in the Key Foods algorithm as only those nutrients included in the Food and Nutrient Database for Dietary Surveys (FNDDS), 5.0 (USDA-ARS, 2012) can be used. The Key Foods approach has allowed NDL to concentrate analytical resources on those foods that contribute significant amounts of nutrients of public health interest to the diet.

Evaluate existing data for scientific quality

At the initiation of NFNAP in 1997, the food composition values in SR were reviewed for scientific quality by NDL staff. Data for many of the foods in the database at that time were found to be more than 10 years old, based on a limited number of values, lacking in complete and accurate documentation, and including some samples of uncertain origin. To assess the quality of existing data and to improve the level of documentation, NDL scientists developed an expert system for evaluating data quality (Holden *et al*., 2002; Holden *et al*., 2005). The expert system focuses on evaluation and documentation of five data quality indicators: 1) sampling plan; 2) sample handling; 3) number of samples analyzed; 4) analytical methodology; and 5) analytical quality control. This system has been used in the production of a number of special interest databases including isoflavones (USDA, 2008), choline (USDA, 2004a), proanthocyanidins (USDA, 2004b), fluoride (USDA, 2005), and flavonoids (USDA, 2007). This process is used to provide information on the data quality assessment for all of the analytical nutrient profiles for foods in SR. Many of the food profiles in the database lacked some or all of the data quality information. For these reasons, and to establish a core set of data of known sampling, analytical methodology, and quality control, NDL determined that comprehensive updates of the food items on the Key Foods list would be needed.

Devise and implement a probability-based sampling survey of US foods

A probability-proportional-to-size (PPS) food sampling plan was developed by NDL staff in collaboration with statisticians from the National Agricultural Statistics Service, USDA (Pehrsson *et al*., 2000). This approach allows the development of nationally representative data for a given food. The original NFNAP food product sampling design was based on a stratified design including each of four regions across the 48 conterminous states of nearly equal in population size. A revised PPS sampling design was developed with 2000 US Census data (Perry *et al.*, 2003) and was based on a stratified three-stage design using 2001 Census Bureau projected state population sizes and Census regions (US Census Bureau, 2002). Forty-eight geographically dispersed counties were selected at the first stage, supermarket outlets at the second stage, and specific food products at the third stage. Subsets of these locations can be selected according to the requirements of the specific food item and nutrients, weighing variability vs. reliability. Multiples of these geographic locations can also be employed

for studies requiring more samples, i.e., where wide variability in a nutrient is expected and/or existing data are limited or nonexistent. Fluoride, for example, is highly variable in drinking water; in a national USDA study, drinking water was sampled in 144 locations and over two seasons (Pehrsson *et al.,* 2006). Another consideration in designing the sampling strategy was that fewer samples would be analyzed for lower consumption foods as identified during the Key Foods process or for nutrients in foods which were not significant contributors to the diet or present in low or trace concentrations. Details of the sampling design are discussed in Perry *et al*. (2003). A new sampling plan based on data from the 2010 US Census has been developed (Perry *et al.*, 2013) and will be used for future sample pickups.

Specific food products were selected according to a sampling approach based on market share. For example, after examining the Key Foods list, it was determined that pizza was a major contributor of many nutrients. Since both pizzas purchased from a fast food pizza restaurant vs. those purchased frozen and heated and served at home are commonly consumed, NDL undertook the analysis of both types. Several different types (e.g., cheese, pepperoni, pepperoni and sausage, and meat/vegetable combinations) and brands (e.g., major national brands and store brands) were purchased in supermarkets as described above. Later, several different types (e.g., cheese, pepperoni, and deluxe) of fast food restaurant pizza from major national chains were purchased from individual restaurants. For frozen pizzas, national composites of each type and brand were prepared. For the fast food restaurant pizzas, four composites of three randomly drawn samples of each type and brand were prepared.

Foods were purchased under contract by a USDA-directed professional product pickup company using tested selection protocols in retail outlets. The foods were shipped to the Food Analysis Laboratory Control Center (FALCC) at Virginia Tech in Blacksburg, Virginia for sample preparation. Procedures were developed for sample unit receipt, preparation, and storage which can be modified as needed for new food samples. FALCC continuously develops protocols for homogenizing and compositing samples based on instructions from NDL. FALCC also collects relevant weights and dissection information for edible and non-edible portions as required, and documents processing and preparation procedures. Processed samples are shipped to USDA-qualified analytical laboratories for analysis as directed by NDL. Reserve and archive samples of each food are maintained at FALCC.

The sampling plan can be modified to meet the requirements of a specific study of specific nutrients or unique foods, e.g., the sampling of tap water in homes to determine fluoride levels. With modifications, the sampling plan can be used for special population groups located in geographically distinct areas (e.g., American Indians and Alaska Natives on reservations, and Hispanic Americans (Perry *et al.,* 2002)).

Analyze sampled foods under USDA-supervised laboratory contracts

NDL employs a two-step process for awarding contracts for analysis of foods. The first step requires prospective contractors to submit a formal proposal. Prospective